INSTRUCTOR GUIDE · PY-STEM · ACTIVITY PYS-07

Pinhole Precision Challenge

The sharpest image, no lens

At a glance: Build the sharpest camera image with no lens. Plan about 80 minutes for the core block; 4 to 6 teams of 4 to 5 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: optics, aperture, straight-line travel of light, camera obscura (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 25 STUDENTS AND 6 TEAMS

- cardstock
- foil
- tape
- pushpins or paper clips
- white screen paper
- light source

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

1. Set up one station per team with the materials above, sorted and ready.
2. Stage the scoring sheet and any reference values before testing begins.
3. Put out PPE (gloves, goggles, trays) before any materials are handled.
4. Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook 0:00 - 0:05

Pose the mission as a challenge: "Build the sharpest camera image with no lens." Take a quick prediction by show of hands.

Phase 2 Prediction 0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints 0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Build or solve 0:18 - 0:44

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Test and record

Teams run the standard test and log a baseline result before changing anything.

Phase 6 Redesign

Each team changes exactly one variable, predicts the effect, and re-tests to compare against baseline.

Phase 7 Score, debrief, cleanup last 12 min

Teams submit a score slip, answer a debrief question with evidence, and reset the station. PPE off, wash or wipe down.

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: No measurable result on the first test. **Fix:** Check the setup against the steps, confirm the standard test was followed, and log the baseline anyway.

Problem: Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to light travels in straight lines.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Why is the image upside down?
- What did shrinking the hole do, and why?
- How is this related to your eye?

SCORING RUBRIC (100 POINTS)

Scoring criterion	Points
Image clarity	35
Concept explanation	25
Build quality	20
Redesign	10
Speed	10
Total	**100**

CLEANUP AND SOURCES

Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

Sources: Exploratorium Personal Pinhole Theater; NASA JPL Pinhole Camera